# Using the Law of Conservation of Energy to solve problems

## Worked solutions + full notes

Working standard: equation first → rearrange → substitute → answer.

## Conservation of energy

Energy cannot be created or destroyed. For the selected system, account for all relevant energy before and after.

**Total energy before = Total energy after**

Task 1.1: Gravitational → Kinetic — falling ball.

Task 1.2: Kinetic → Gravitational — coasting car uphill.

Task 1.3: Gravitational → Internal — lift descending steadily.

Task 1.4: Elastic → Kinetic — spring launcher.

Task 1.5: Kinetic → Internal — braking bicycle.

Task 1.6: Kinetic → Elastic — bouncy ball during compression; some energy also goes to internal/sound.

## Task 2 · Simple conservation

A Equation: $E_{electrical} = \Delta U_{int} + E_{radiation}$

Substitution: $100 = 75 + E_{radiation}$

Answer: 25 J each second.

B Equation: $|\Delta U_g| = \Delta K + E_{other}$

Substitution: $250 = 200 + E_{other}$

Answer: 50 J mainly to internal energy.

C Equation: $U_{s,i} + K_i = U_{g,f} + K_f$

With $K_i=K_f=0$, $U_{s,i}=\Delta U_g=25$ J.

## Task 3 · Falling ball

$m=0.028$ kg, $\Delta y=-0.78$ m.

Equation: $\Delta U_g=mg\Delta y$

Substitution: $(0.028)(9.8)(-0.78)$

Answer: $-0.214$ J $\approx -0.21$ J.

Equation: $v^2=v_0^2+2a\Delta y$

Substitution: $v^2=0+2(-9.8)(-0.78)$

Answer: 3.9 m $s^{-1}$.

Equation: $K=\frac{1}{2}mv^2$

Substitution: $\frac{1}{2}(0.028)(3.91^2)$

Answer: 0.21 J.

## Task 5 · Bouncy ball practical

Equation: $U_{g,i}=mgh_1$

Equation: $U_{g,rebound}=mgh_2$

Equation: $\Delta E_{mech,not\ recovered}\approx mg(h_1-h_2)$

This unrecovered mechanical energy is mainly transferred to internal energy and sound.

## Task 6A · Incline

Equation: $\sin\theta=0.30/0.50 \rightarrow \theta=36.9°$.

Equation: $F_{g,\parallel}=F_g\sin\theta=mg\sin\theta$

Substitution: $(1.2)(9.8)(0.60)$

Answer: 7.1 N.

Equation: $\Sigma F_{\parallel}=F_{net}=ma$

Substitution: $7.056=(1.2)a$

Answer: 5.9 m $s^{-2}$.

Equation: $v^2=v_0{}^2+2a\Delta x$

Answer: 2.4 m $s^{-1}$.

Equation: $\Delta U_g=mg\Delta y$

Answer: −3.5 J.

Equation: $K=\frac{1}{2}mv^2$

Answer: 3.5 J.

## Task 6B · Projectile

Equation: $v_{x,0}=v_0\cos\theta \rightarrow 13.0$ m $s^{-1}$.

Equation: $v_{y,0}=v_0\sin\theta \rightarrow 7.5$ m $s^{-1}$.

Equation: $v_y{}^2=v_{y,0}{}^2+2a_y\Delta y \rightarrow \Delta y=2.87$ m.

Equation: $K=\frac{1}{2}mv^2 \rightarrow K_i=28.1$ J.

At top $v_y=0$ and $v_x=13.0$ m $s^{-1}$.

Equation: $K=\frac{1}{2}mv^2 \rightarrow K_{top}=21.1$ J.

Equation: $\Delta K=K_f-K_i \rightarrow -7.0$ J.

Equation: $\Delta U_g=mg\Delta y \rightarrow +7.0$ J to top; −29.4 J roof→ground.

Equation: $K_i+U_{g,i}=K_f+U_{g,f} \rightarrow K_{ground}=57.5$ J.

Equation: $K=\frac{1}{2}mv^2$; rearranged: $v=\sqrt{(2K/m)} \rightarrow 21.5$ m $s^{-1}$.

## Task 7 · General gravitational potential

Equation: $U_G = -Gm_1m_2/r$

At $R_E$: -4.064e+09 J ≈ $-4.1\times10^{9}$ J.

Equation: $U_G = -Gm_1m_2/r$

At $2R_E$: -2.032e+09 J ≈ $-2.0\times10^{9}$ J.

Equation: $\Delta U_G = U_{G,f} - U_{G,i} \rightarrow +2.0\times10^{9}$ J.

Escape to infinity: $U_{G,f} = 0$, so required energy ≈ $4.1\times10^{9}$ J.

Near-Earth equation: $\Delta U_g = mg\Delta y \rightarrow 4.1\times10^{9}$ J over 6380 km; invalid because g changes strongly.

## Task 8 · Ball with drag

Equation: $|\Delta U_g| = mgh$

Answer: 71.3 J ≈ 71 J.

Equation: $|\Delta U_g| = \Delta K \rightarrow K_f = 71$ J.

Equation: $K = \frac{1}{2}mv^2$; rearranged: $v = \sqrt{(2K/m)} \rightarrow 17$ m $s^{-1}$.

Equation: $|\Delta U_g| = K_f + \Delta U_{int}$; half to internal → $K_f = 35.7$ J.

Equation: $K = \frac{1}{2}mv^2 \rightarrow 12$ m $s^{-1}$.

Equation: $v^2 = v_0{}^2 + 2a\Delta x \rightarrow a = 4.9$ m $s^{-2}$ downward.

Equation: $\Sigma F = F_{net} = ma$

Force equation: $F_g - F_{drag} = ma \rightarrow F_{drag} = 2.5$ N upward.

## Task 9 · Derivations

Equation: $K_i+U_{g,i}=K_f+U_{g,f}$

From rest: $MgH=\frac{1}{2}Mv^2 \rightarrow v=\sqrt{(2gH)}$.

With initial $v_0$: $\frac{1}{2}Mv_0^2+MgH=\frac{1}{2}Mv^2 \rightarrow v^2=v_0^2+2gH$.

## Task 10 · Projectiles

Horizontal launch equations: $K=\frac{1}{2}mv^2$; $\Delta U_g=mg\Delta y$; $\Delta K+\Delta U_g=0$.

$K_i=31.25$ J; $\Delta U_g=-294$ J; $K_f=325.25$ J; $v=16.1$ m $s^{-1}$.

General equation: $K_i+U_{g,i}=K_f+U_{g,f} \rightarrow v=\sqrt{(v_0^2+2gH)}$.

Angled launch: $K=\frac{1}{2}mv^2 \rightarrow K_i=36$ J; $v_x=v_0\cos\theta=6.0$ m $s^{-1}$; at top $K=9.0$ J; $\Delta K+\Delta U_g=0$ gives $\Delta U_g=27$ J; $\Delta U_g=mg\Delta y$ gives $h=5.5$ m.

## Task 11 · Incline

Equation: $F_{g,\parallel}=F_g\sin\theta=mg\sin\theta \rightarrow 9.8$ N.

Equation: $\Sigma F_{\parallel}=F_{net}=ma \rightarrow a=4.9$ m $s^{-2}$.

Equation: $v^2=v_0^2+2a\Delta x \rightarrow v=1.98$ m $s^{-1}$.

Equation: $h=d\sin\theta \rightarrow h=0.20$ m.

Equation: $K_i+U_{g,i}=K_f+U_{g,f} \rightarrow K_f=mgh=3.92$ J.

Equation: $K=\frac{1}{2}mv^2 \rightarrow v=1.98$ m $s^{-1}$.

General: $Mgd\sin\theta=\frac{1}{2}Mv^2 \rightarrow v=\sqrt{(2gd\sin\theta)}$.

## Task 12 · Rough incline algebra challenge

(a) Equation: $\sin\theta = h/d \rightarrow d = h/\sin\theta$.

(b) Equation: $\Sigma F_{\perp} = 0 \rightarrow F_n - F_g\cos\theta = 0 \rightarrow F_n = mg\cos\theta$.

(c) Equation: $F_f = \mu_k F_n \rightarrow F_f = \mu_k mg\cos\theta$.

(d) Equation: $\Sigma F_{\parallel} = F_{net}$; $F_{net} = F_g\sin\theta - F_f \rightarrow mg(\sin\theta - \mu_k\cos\theta)$.

(e) Equation: $F_{net} = ma \rightarrow a = g(\sin\theta - \mu_k\cos\theta)$.

(f) Equation: $v^2 = v_0^2 + 2a\Delta x \rightarrow v = \sqrt{[2gh(1 - \mu_k\cot\theta)]}$.

(g) Equation: $K = \tfrac{1}{2}mv^2 \rightarrow K_f = mgh(1 - \mu_k\cot\theta)$.

(h) Equation: $\Delta U_g = mg\Delta y \rightarrow -mgh$.

(i) Equation: $K_i + U_{g,i} + U_{int,i} = K_f + U_{g,f} + U_{int,f} \rightarrow \Delta U_{int} = \mu_k mgh\cot\theta$.